Predictive vs Generative

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

TEACHER GUIDE · 1 OF 9

Predictive vs Generative

> Prepare Predictive vs Generative: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Distinguish discriminative **prediction** from generation by inspecting the requested **output**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/27, ai-3/1. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Image and Voice Pipelines

Without opening the old lesson, explain one key idea from Image and Voice Pipelines. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**

Reference example, not the only acceptable wording: Teacher explanation: A multimodal pipeline may transcribe speech, inspect an image and generate a response.

Errors at one stage can propagate even when later stages run correctly. Track source modality, segment/time references, transcription, interpretation and final claims separately.

Compare conflicting **evidence** without assuming one modality always wins. Ask for clarification when the task needs unresolved information.

Feedback Loops

Without opening the old lesson, explain one key idea from Feedback Loops. Give an example and a mistake that the idea helps you avoid.

Look for: Implement a deterministic recommender

Reference example, not the only acceptable wording: Both score 8 and qualify; A is alphabetically first under the tie rule.

Similarity by Selected Traits

Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

Look for: Represent items with numeric features

Reference example, not the only acceptable wording: Energy 4, group work 1.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A tool retrieves a book description, predicts a genre, then writes a new blurb. Identify each operation and give one distinct failure to check at each stage.

Concept to explain

Predictive systems select a target; **generative** systems construct new **output**.

Term	Meaning
predictive	Estimating a target from supplied inputs.
generative	Producing new content from learned patterns and supplied conditions.
output space	The set or range of outputs a system can produce for a task.
fit for purpose	Suitable for the stated users, task, conditions and consequences.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Use two 45-minute sessions. All model **output**s and probability tables are constructed classroom examples. No hosted AI service, external API or student account is needed.

Distinguish a model objective from factual support and system permissions. Collect independent work before discussing its solutions.

Readiness answer

A **prediction** is a model **output**. Truth requires appropriate independent **observation**s or sources; confidence and low **training** loss do not supply that **evidence**.

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Prepare the independent task and fresh retry without displaying solutions

End goal for the pictorial case

By the end, I can distinguish a generated continuation from a **prediction** checked against the world.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Predictive vs Generative using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. ‘Is this book suitable?’ → classification.

2. ‘How many days to finish?’ → numeric **prediction**.

3. ‘Write a spoiler-free teaser’ → generation.

4. A product may combine them, but each stage needs its own test.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

> **The teaching case**
>
> A labels genre; B writes a new blurb; C computes 24/6.

> **The teaching case**
>
> Distinguish operations.

Exact explanation to model

> **Why the result follows**
>
> A predicts a category, B generates text, C calculates 4. Products may combine them.

Explain the added worked example

Explain the added worked example · 1

A classifier might map a message to the label request or report. A text generator instead produces a sequence, predicting a distribution over the next token given a prefix.

Both make **prediction**s, but the **training** targets and permitted **output**s differ. One application can combine generation with a classifier, retrieval and human approval.

Calling a whole product "**generative**" does not describe every component.

The objective defines what **training** attempts to fit. Next-token fit rewards matching observed continuations; it does not directly certify that an answer is true, well sourced or permitted.

A fluent completion is still an **output** to evaluate against the task.

Explain the added worked example · 2

There are ten observed continuations. The empirical probabilities are red 6/10 = 0.6, blue 3/10 = 0.3, green 1/10 = 0.1. They are nonnegative and sum to 1.

A simple frequentist classroom model uses these relative counts as its next-token distribution. These probabilities describe the supplied corpus counts, not the current colour of any actual marker.

Key terms

Term	Meaning
Generative objective	A training aim concerned with producing or modelling possible outputs
Discriminative objective	A training aim mapping inputs to a specified target such as a label
Conditional distribution	Probabilities for alternatives given stated information
Autoregressive generation	Producing a sequence one next item at a time using preceding items

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Model the reasoning aloud · 3

After producing red, the next step conditions on the expanded prefix “The marker is red”. A second table might give an end token probability 0.8 and the token today probability 0.2.

A generated sequence can stop at the end token or continue. Do not reuse the first-step table unchanged after adding a token unless the model explicitly ignores the new prefix.

Model the reasoning aloud · 4

A separate classifier can return red as a label from an image; that is a different **input** and target from continuing text. A downstream evaluation asks whether the system correctly reports the current marker.

It must include ground truth or sources and appropriate failure handling, not just count plausible phrases.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Predictive vs Generative

A fictional toy generator predicts the next word after “The box is”.

The facts we will use

Supplied facts

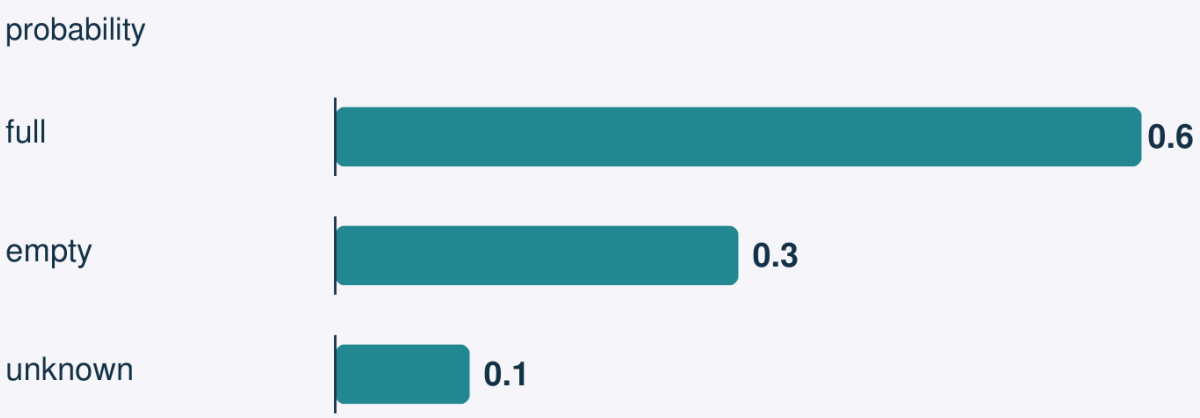
Observed **training** continuations: full 6, empty 3, unknown 1.

Supplied facts

Use counts divided by total 10. The current physical box is separately checked and found empty.

AI ACADEMY / AI-2/27

Toy next-word distribution



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/27

Predictive vs Generative

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count table

Continuation counts are 6,3,1; total 10.



2. Probabilities

full=0.6; empty=0.3; unknown=0.1.



3. Greedy continuation

Largest probability chooses full for this toy prefix.



4. Check reality

Current observation says empty; frequent wording is not a present inventory reading.

CHANGE ONE THING

Change the training table to full 2, empty 7, unknown 1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Count table	Continuation counts are 6,3,1; total 10.
2. Probabilities	full=0.6; empty=0.3; unknown=0.1.
3. Greedy continuation	Largest probability chooses full for this toy prefix.
4. Check reality	Current observation says empty; frequent wording is not a present inventory reading.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The table predicts a continuation in a text exercise, not the contents of the current box. Generating the most frequent token is a model-defined decision. A factual inventory statement needs current **evidence**.

The numerical table is intentionally small; real **generative** systems condition on richer context and do not simply consult these three counts.

What this example does not establish

Keep the stated limit

A one-step toy distribution does not describe every **predictive** or **generative** model or validate factual accuracy.

Change one thing in the pictured case

Change one thing

Change: Change the **training** table to full 2, empty 7, unknown 1.

Predicted result

Prediction: Greedy **output** becomes empty with probability 0.7.

Why the result follows

Reason: The continuation distribution changed; the independently observed box has not changed.

Ask while showing the picture

Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Predictive vs Generative, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A labels genre; B writes a new blurb; C computes 24/6.

2. State the decision to explain

Write the question to explain: Distinguish operations.

3. Reveal the worked reasoning

Reveal after **prediction**s: A predicts a category, B generates text, C calculates 4. Products may combine them.

Hold back the independent task

Keep for an independent attempt: One model labels a review positive or negative; another writes a new review. Which classifies and which generates? Must generated text be true?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: open 0.8 and closed 0.2. The corpus prefers open, but the sensor supplies current closed **evidence**. Exit: a conditional continuation probability is not verification of a current fact.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and distinguish a numerical claim from its interpretation.
7 to 18 minutes	Read section 1 and complete W1; ask for a reason alongside each answer.
18 to 32 minutes	Trace the worked case in section 2 and complete W2.
32 to 45 minutes	Discuss the diagram and complete W3 in pairs; record support.
Session 2 start to 8 minutes	Complete the guided response before sharing the answer.
8 to 20 minutes	Read the remaining sections and trace another constructed case.
20 to 35 minutes	Collect W4 independently, requiring calculations and a bounded conclusion.
35 to 45 minutes	Discuss W5 and the exit response. Provide feedback and a fresh retry.

Reduce help deliberately

Completion practice

Completion task: For The marker is, calculate the three empirical probabilities, their sum and their meaning.

Reasoning to listen for

Teacher check: Teacher explanation: Probabilities 0.6,0.3,0.1 sum to 1 and describe ten supplied corpus continuations; they are not **observation**s of the current marker.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Generative** and **predictive** objectives. Proposed explanation: "**Generative** models do not predict". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: **Autoregressive generation** predicts the next item repeatedly.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

> Does matching the physical box in the changed table prove the generator measures boxes?

Reasoned teacher answer

No. It happens to match here; its **mechanism** still selects a token from supplied counts rather than sensing contents.

Changed-case reasoning: Greedy **output** becomes empty with probability 0.7. The continuation distribution changed; the independently observed box has not changed.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Predictive vs Generative investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Sort nine task cards by **mechanism**, name the **output space**, and specify one evaluation method for each category.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Generative models do not predict: **Autoregressive generation** predicts the next item repeatedly.

The most frequent phrase is the current truth: Corpus frequency and current **observation**s answer different questions.

A single accuracy score evaluates the whole system: Generation, retrieval, grounding, formatting and permissions require appropriate separate checks.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Predictive vs Generative questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

A predicts a category, B generates text, C calculates 4. Products may combine them.

Independent question

Independent question · withhold the key

One model labels a review positive or negative; another writes a new review. Which classifies and which generates? Must generated text be true?

Expected independent answer

Teacher key · Expected independent answer

The first classifies; the second generates. Generated text can be inaccurate or invented, so a fluent review is not verified **evidence**.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Generating text proves the model checked the facts.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Retrieval may fetch the wrong edition; **prediction** may choose the wrong genre; generation may invent an unsupported detail. Check document identity, the genre against reliable **evidence** and each factual claim in the blurb.

Fluency does not validate the preceding stages.

Explanations for the additional workbook examples

W1 Identify the objective

Workbook question · W1

Classify these components by target: label a message request/report; continue a prefix with a sequence; retrieve a stored sensor record. Explain why the application may combine them.

Teacher reasoning · W1

Teacher explanation: Message labels are discriminative **prediction**; continuing a sequence is generation; record lookup is retrieval rather than learned generation by itself.

One application can route a request, retrieve **evidence** and generate a response.

W2 Compute the distribution

Workbook question · W2

For The marker is, calculate the three empirical probabilities, their sum and their meaning.

Teacher reasoning · W2

Teacher explanation: Probabilities 0.6,0.3,0.1 sum to 1 and describe ten supplied corpus continuations; they are not **observation**s of the current marker.

W3 Follow the prefix

Workbook question · W3

After red is selected, the expanded-prefix table gives end 0.8 and today 0.2. Explain why these values cannot be substituted for the red/blue/green probabilities.

Teacher reasoning · W3

Teacher explanation: The conditional prefix and alternatives differ. The second table describes what follows The marker is red, not what colour follows the earlier prefix.

W4 Analyse a new corpus independently

Workbook question · W4

For prefix The signal is, counts are high 2, low 5, unknown 3. Compute all probabilities and the most frequent continuation. A current measurement reports high.

State what the most frequent continuation does and does not establish.

Teacher reasoning · W4

Teacher explanation: Probabilities high 0.2, low 0.5, unknown 0.3; low is most frequent. This describes the constructed corpus, not the current signal.

Current high is supported by the supplied measurement; generating low would conflict with that **observation**.

W5 Design a useful evaluation

Workbook question · W5

A system reports current room status using retrieved measurements and generated sentences. Name checks for **evidence**, response quality and permission.

Teacher reasoning · W5

Teacher explanation: Check the correct room and measurement time/source, agreement of every status claim with retrieved data, required response format and handling of missing/conflicting **observation**s, and that only approved read/report actions occur.

Continuation fit alone is insufficient.

Assess independent transfer

Independent question · withhold the key

Selected case: For prefix The signal is, counts are high 2, low 5, unknown 3. Compute all probabilities and the most frequent continuation. A current measurement reports high.

State what the most frequent continuation does and does not establish.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: Probabilities high 0.2, low 0.5, unknown 0.3; low is most frequent. This describes the constructed corpus, not the current signal.

Current high is supported by the supplied measurement; generating low would conflict with that **observation**.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

Connect Predictive vs Generative to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

Keep scoring **predictive** and explanations templated or carefully generated and verified.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 28: Prompt Experiments. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide labelled intermediate columns and allow a calculator. Ask students to state the meaning and denominator of a value before interpreting it.

Extension: implement the classroom calculation locally and test boundary cases without using personal data.

W4 requires 0.2,0.5,0.3 and low, with the corpus/current-**evidence** distinction. Fresh retry: counts ready 7, busy 1, unknown 2 and current **observation** busy.

Probabilities 0.7,0.1,0.2; ready is most frequent but does not establish the current state.

Week 2 defines a small token vocabulary and traces encoding and decoding without assuming tokens are words.

Plan the delayed return

Suggested return points in this level: ai-2/28, ai-2/30, ai-2/33. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Use approved book facts for recommender explanations; generated fluency is not a replacement for catalogue **evidence**.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Predictive vs Generative lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Distinguish discriminative **prediction** from generation by inspecting the requested **output**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: For The marker is, calculate the three empirical probabilities, their sum and their meaning.

Supported practice answer

Reasoned practice answer: Teacher explanation: Probabilities 0.6,0.3,0.1 sum to 1 and describe ten supplied corpus continuations; they are not **observation**s of the current marker.

Check every learner before discussion: Use the worked case for **Generative** and **predictive** objectives. Proposed explanation: “**Generative** models do not predict”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: **Autoregressive generation** predicts the next item repeatedly.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Change the **training** table to full 2, empty 7, unknown 1. Require a **prediction** and a reason before comparing. Expected result: Greedy **output** becomes empty with probability 0.7.

Reason: The continuation distribution changed; the independently observed box has not changed.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

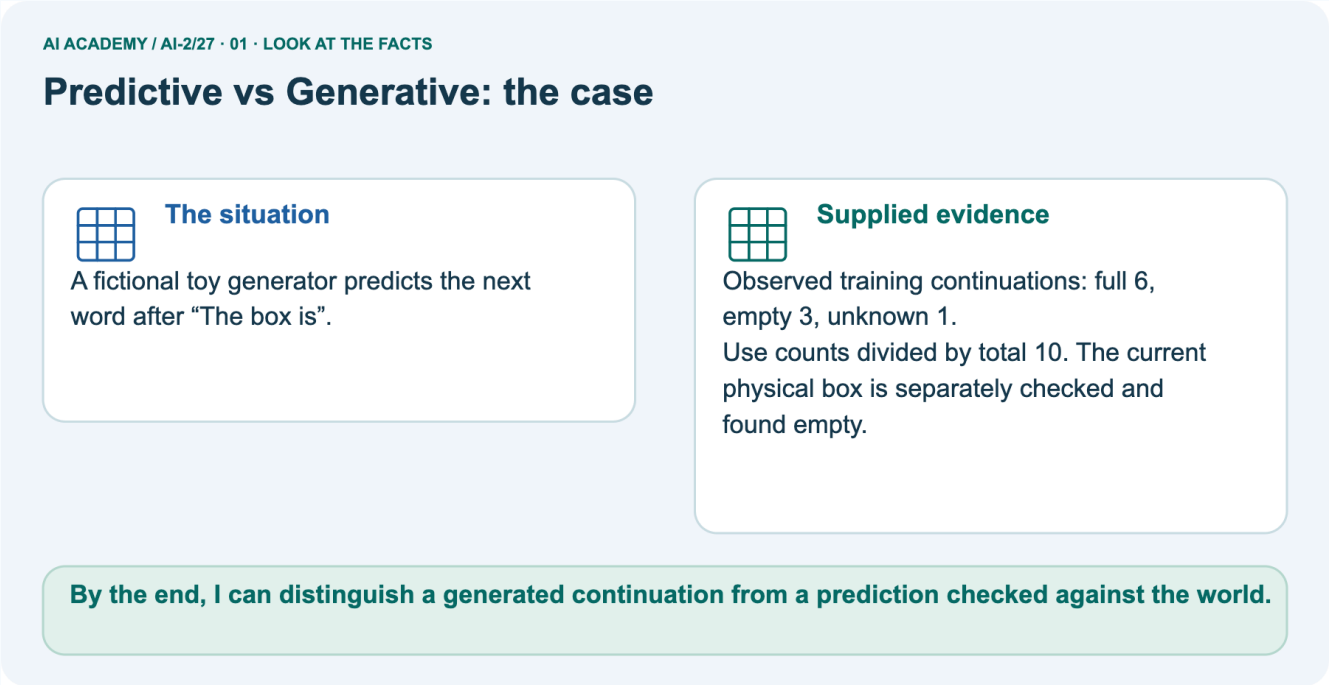
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Predictive vs Generative graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Predictive vs Generative: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional toy generator predicts the next word after “The box is”. Supplied facts: Observed **training** continuations: full 6, empty 3, unknown 1. Use counts divided by total 10.

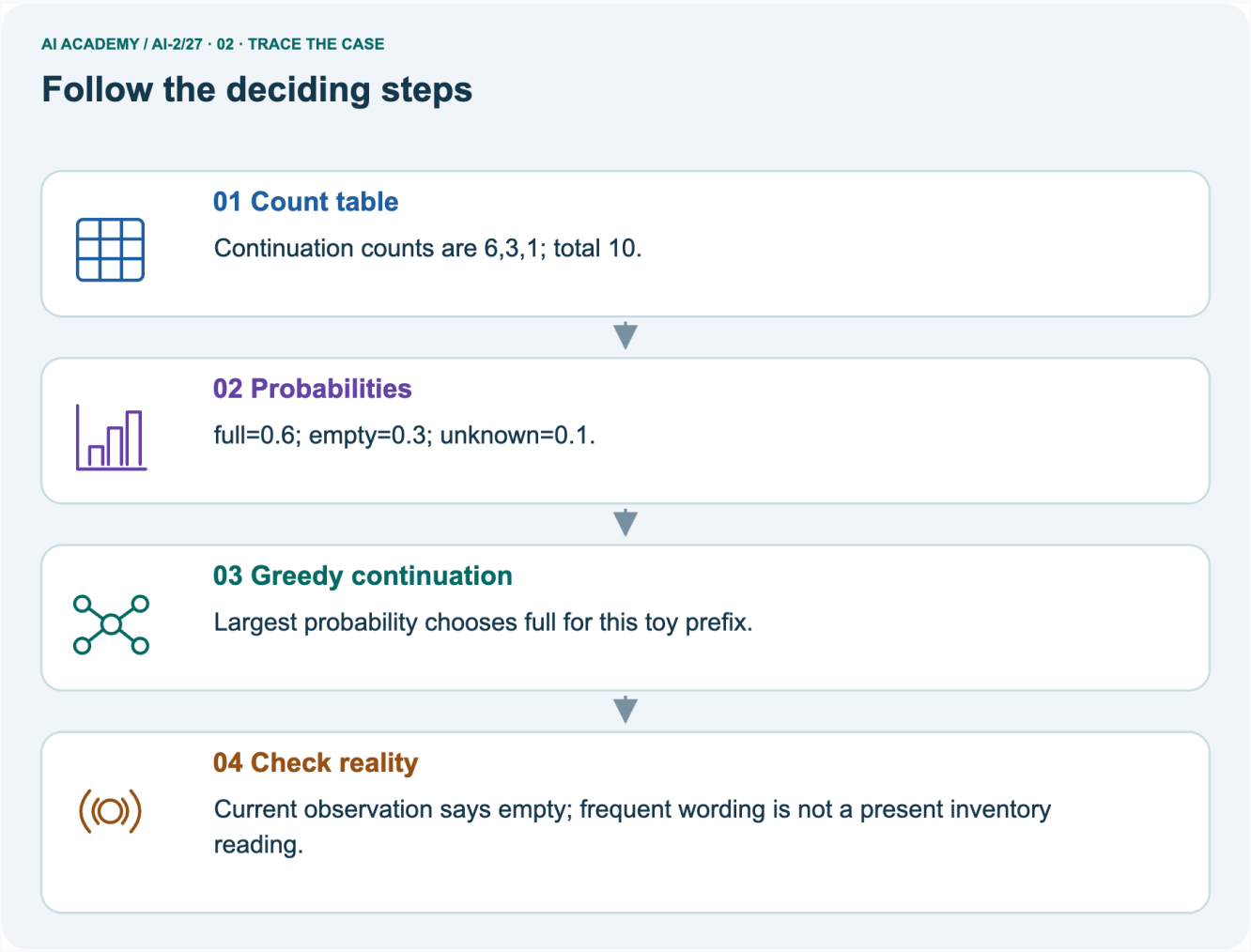
The current physical box is separately checked and found empty.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

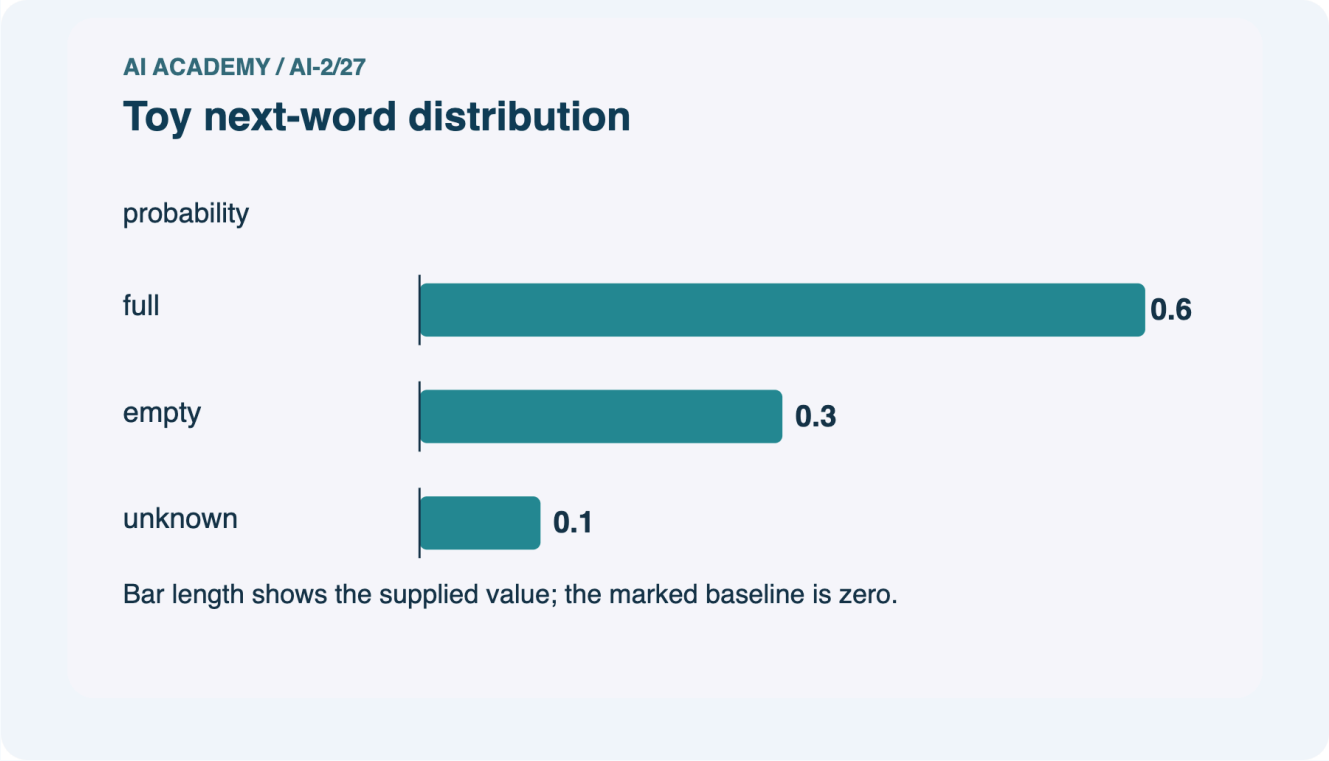
Count table: Continuation counts are 6,3,1; total 10. → Probabilities: full=0.6; empty=0.3; unknown=0.1. → Greedy continuation: Largest probability chooses full for this toy prefix. → Check reality: Current **observation** says empty; frequent wording is not a present inventory reading.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Toy next-word distribution



Follow the labelled picture with the matching explanation. Open the vector diagram

Toy next-word distribution. Units: probability. full: 0.6. empty: 0.3. unknown: 0.1.

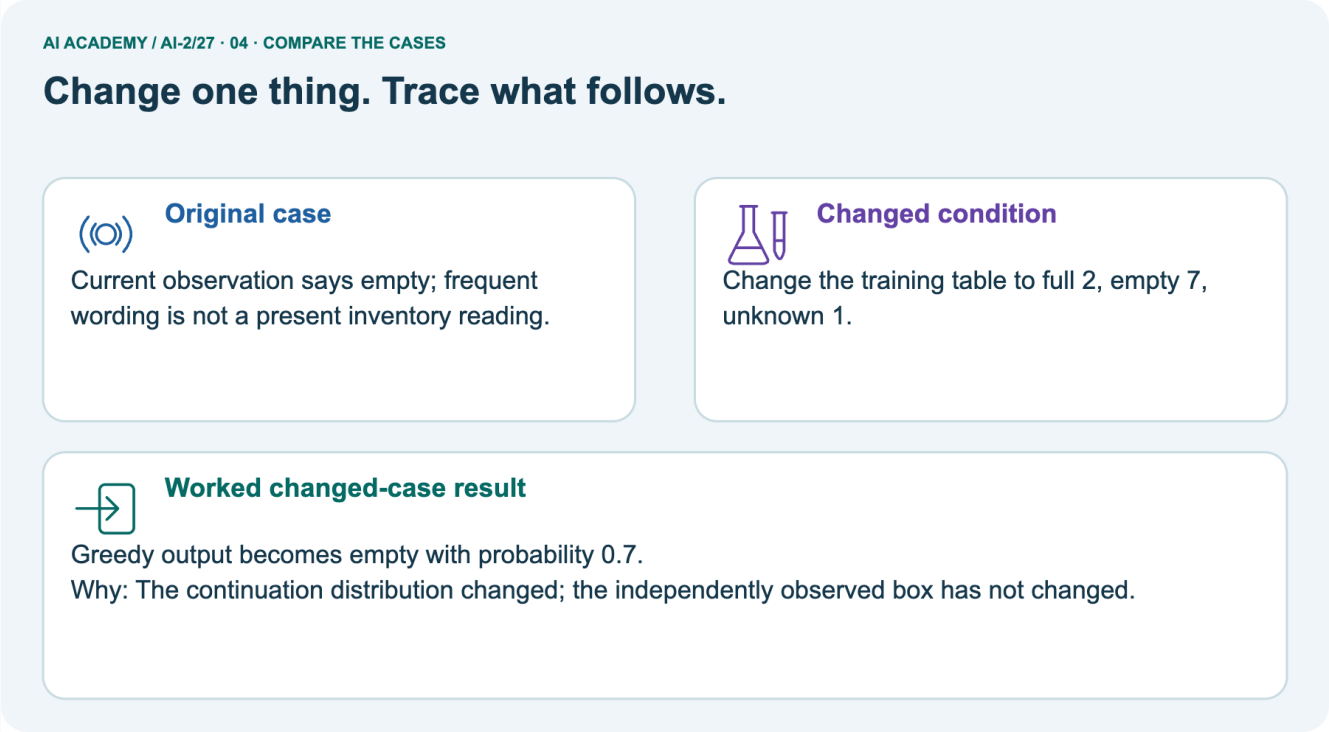
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Current **observation** says empty; frequent wording is not a present inventory reading. Changed condition: Change the **training** table to full 2, empty 7, unknown 1.

Expected result: Greedy **output** becomes empty with probability 0.7. Reason: The continuation distribution changed; the independently observed box has not changed.

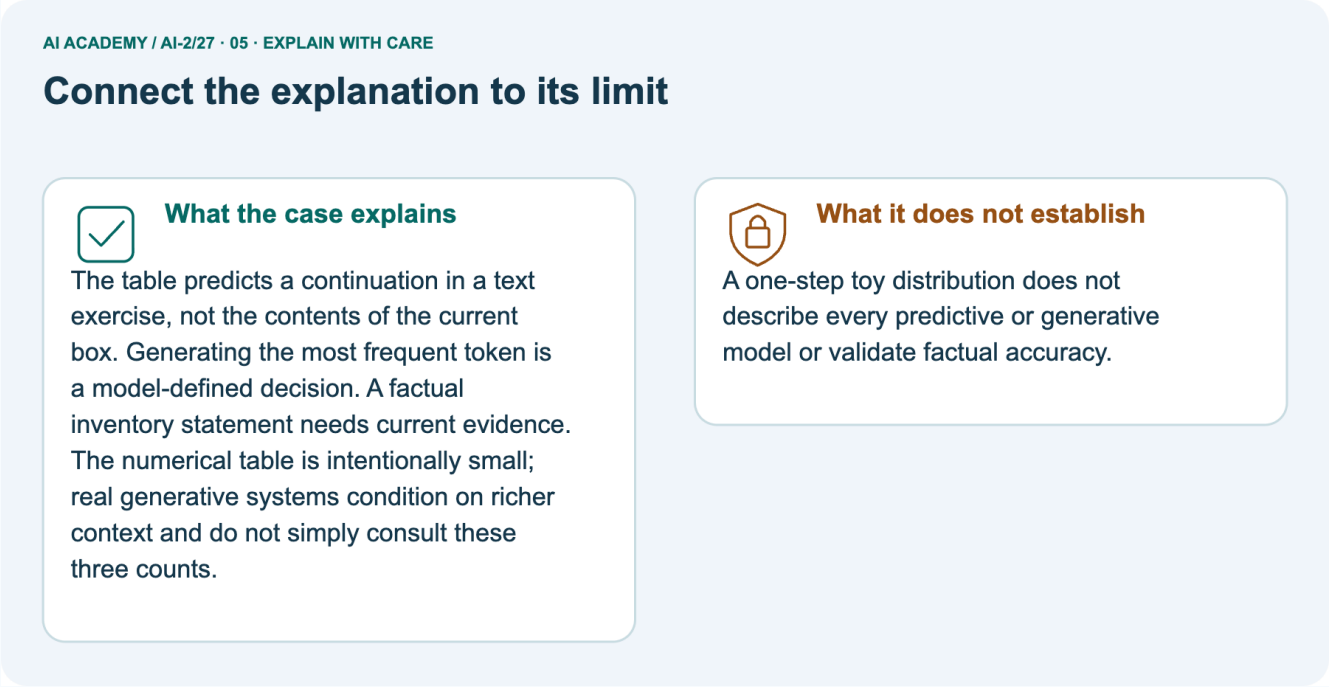
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The continuation distribution changed; the independently observed box has not changed.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The table predicts a continuation in a text exercise, not the contents of the current box. Generating the most frequent token is a model-defined decision. A factual inventory statement needs current **evidence**.

The numerical table is intentionally small; real **generative** systems condition on richer context and do not simply consult these three counts.

Limit: A one-step toy distribution does not describe every **predictive** or **generative** model or validate factual accuracy.

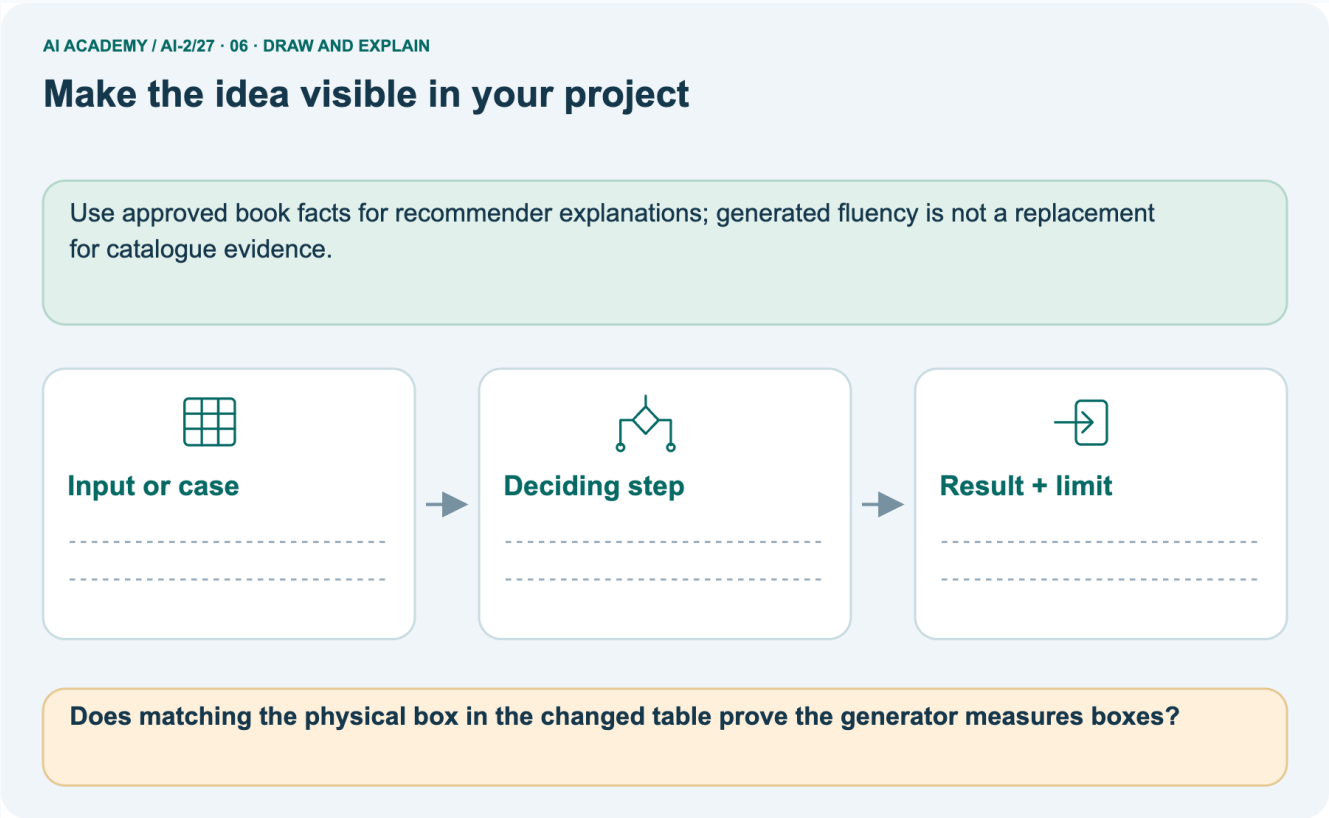
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A one-step toy distribution does not describe every **predictive** or **generative** model or validate factual accuracy.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use approved book facts for recommender explanations; generated fluency is not a replacement for catalogue **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: Does matching the physical box in the changed table prove the generator measures boxes?

> Draw your project trace and justify the deciding step using **evidence**.

> **Discuss while showing the picture**
>
> Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: No. It happens to match here; its **mechanism** still selects a token from supplied counts rather than sensing contents.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.